

EDUCATION

Georgia State University – Atlanta, GA Bachelor of Science in Computer Science | Expected April 2026

- Relevant Coursework: Data Structures, Algorithms, Systems Programming, Operating Systems, Software Engineering, Software Development, Data Science Fundamentals, Calculus I and II, Linear Algebra, Statistics, Mobile Development, Web Programming, Cybersecurity Fundamentals

Certificates of Git Essential Training: The Basics | September 2025

WORK EXPERIENCE

Freelance

Software Engineer | Georgia | June 2024 – Present

- Developed **6+ full stack web applications** for local businesses using **HTML, CSS, JavaScript, PHP, and SQL** delivering scalable, user-friendly, and production ready software.
- Deployed websites to production, using **Vercel** hosting, **MySQL** for database configuration, **Git** based version control, and domain setup.
- Improved product usability, accessibility, and performance, directly contributing to an average of **20%** increase in user engagement through client projects.
- Gathered requirements through conversations with clients and translated business needs to technical solutions demonstrating strong communication and product thinking skills.
- Implemented authentication systems, including flows for logging in and creating accounts, password hashing, session management, and input validation to prevent security risks.

Select Management Resources

Software Engineer Intern | Alpharetta, Georgia | January 2023 – July 2023

- Developed **Python** and **JavaScript** scripts to streamline troubleshooting workflows, reduce repetitive operational tasks, and improve incident response efficiency.
- Designed and maintained structured technical documentation, system guides, and process workflows, improving onboarding efficiency by **20%**.
- Investigated application, system, and network issues through log analysis, ticket data review, and root cause analysis, contributing to a **25%** reduction in recurring incidents.
- Collaborated with infrastructure and operations teams to identify system reliability gaps and implement tooling and process improvements.
- Debugged and resolved complex software and system issues using structured problem solving, scripting-based validation, and logical decomposition.
- Produced internal data driven reports and dashboards to track issue trends and performance metrics, enabling data driven operational decisions.

PROJECTS

Traffic Sign Image Recognition System (Python, PyTorch, scikit-learn, OpenCV, Pandas, NumPy, Torchvision, Matplotlib)

- Built an image recognition and classification models for **43** traffic sign classes using the **GTSRB** dataset, training, evaluating, and optimizing multiple machine learning models.
- Developed a **Neural Network** using **ResNet18** and **Transfer Learning** achieving up to a **94%** test accuracy through stage-based training, and fine tuning.
- Implemented multiple baseline models including **HOG + SVC**, **k-NN**, and **Random Forest** models, and through hyperparameter tuning was able to improve model test accuracies from as low as **77%** to as high as **97%** improving overall model accuracies by **10%-20%**.
- Evaluated models based statistical analysis using precision, recall, F1-score, and matplotlib for confusion matrices visualization to analyze class level errors to better tailor optimizations and improve accuracies
- Engineered data preprocessing and augmentation pipelines, including normalization and standardization of the image features.

ML Debug Assist (Python, scikit-learn, YAML)

- Designed and built a fully local ML-powered debugging assistant that classifies Python runtime errors using a hybrid rule-based and machine-learning approach to generate structured troubleshooting guidance.
- Implemented a pipeline covering synthetic dataset generation, text preprocessing, TF-IDF feature extraction, and multiclass Logistic Regression training.
- Added confidence-aware inference logic to dynamically adjust output, surfacing multiple candidate error categories and prompting for additional context when predictions were uncertain.
- Developed YAML-driven fix playbooks with similarity-based retrieval to deliver actionable debugging checklists and contextual examples, enabling reproducible, offline execution via a CLI.

Atlas: Social Fitness & Nutrition Platform (Flutter, Dart, Firebase, REST APIs, AI APIs)

- Developed a large mobile application that combines fitness tracking, nutrition logging, Artificial Intelligence driven planning, and social media functionality.
- Implemented secure authentication, user profiles, posts, likes, comments, and follower relationships using **Firebase Authentication**, **Firestore**, **Storage**, and security rules.
- Integrated multiple third-party REST APIs to support exercise discovery, workout logging, macro tracking, barcode food lookup, and nutrition data retrieval.
- Built an Artificial Intelligence powered assistant using **OpenAI API** to generate personalized workout and meal plans through structured prompt engineering based on user constraints (equipment, dietary preferences, environment).
- Created modular application pages including **Feed**, **Tracking (Workouts & Nutrition)**, **Artificial Intelligence Assistant**, and **Profile** pages with scalable data models and role-based access control.

TECHNICAL SKILLS

- Programming Languages: Python, Java, JavaScript, C++, C, C#, Dart, HTML, CSS, SQL, PHP, YAML
- Technologies: GitHub, Git, Emulators / Simulators, Xcode, Microsoft Office, Google Suite, MySQL, DBeaver, Firebase, Linux, Unix, Command Line, Unit Testing, Data Structures, Algorithms, Flutter
- Core Skills: Object Oriented Programming, Database Management, Full stack Development, App Development, Web Development, Debugging, Troubleshooting, Automation Scripting, UI/UX, Server-Side Programming, Version Control, REST APIs, AI API Integrations, Prompt Engineering, Mobile UI Architecture, Machine Learning Model Development, Data Processing